

Paloma Sodhi

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SUMMARY

Research Scientist with ML and NLP expertise working on dialog agents that converse and act. Specializing in large language models, reinforcement learning, and alignment using AI feedback. Experienced in leading independent research projects, training large-scale models, and deploying them in production.

WORK EXPERIENCE

ML/NLP Research Scientist

May 2022 – Present

ASAPP Research

New York City, NY

- Building ML and NLP methods for customer service chatbots that both converse with customers and take actions.
- Trained LLaMA3-8b, 70b models with AI feedback for dialog agents, tool use, and coding, showing smaller models can outperform GPT-4o when using privileged information.^[1]
- Led development of new LLM agent architectures that decompose complex web tasks into modular policies. Open-sourced models achieving the top result on WebArena leaderboard^[2]. Deployed these in production.^[7]
- Investigated PPO vs offline RL^[3,4] and preference learning methods like DPO, KTO for conversational agents.

Visiting Researcher AI

Mar 2021 – Mar 2022

Facebook AI Research

Pittsburgh, PA

- Developed efficient inverse reinforcement learning methods to learn reward models for large-scale factor graphs.^[6]
- Worked on an open-source differentiable optimizer library for training end-to-end models.^[5]

Graduate Researcher

Jan 2016 – Mar 2022

Carnegie Mellon University

Pittsburgh, PA

- Ph.D. on combining machine learning with graph optimizers in the loop, with applications to robot manipulation.

AI Research Intern

Jun 2020 – Aug 2020

Facebook AI Research

Remote

- Built multimodal vision-tactile models for embodied AI applications like in-hand robot manipulation.

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Ph.D. in Robotics, School of Computer Science, GPA: 4.0 / 4.0

Aug 2017 – Mar 2022

Carnegie Mellon University

Pittsburgh, PA

M.S. in Robotics, School of Computer Science, GPA: 4.06 / 4.0

Aug 2015 – May 2017

Indian Institute of Technology Guwahati

Guwahati, India

B.S. in Electrical Engineering

Aug. 2009 – May 2013

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, HTML/CSS **Libraries:** HF Transformers, PyTorch, Tensorflow, Gym, SB3

SELECTED PAPERS

- [1] S. Choudhury* and **P. Sodhi***. *Better than Your Teacher: LLM Agents that learn from Privileged AI Feedback International Conference on Learning Representations (ICLR) 2025*, in submission. *equal contribution [paper] [code]
- [2] **P. Sodhi**, S. Branavan, Y. Artzi, R. McDonald. *SteP: Stacked LLM Policies for Web Actions. Conference on Language Modeling (COLM) 2024*. [paper] [code]
- [3] **P. Sodhi**, F. Wu, E. Elenberg, K. Weinberger, and R. McDonald. *On the Effectiveness of Offline RL for Dialogue Response Generation. International Conference on Machine Learning (ICML) 2023*. [paper] [code]
- [4] D. J. Min, **P. Sodhi**, R. Ramakrishnan. *Workflow-Guided Response Generation for Task-Oriented Dialogue. arXiv 2023*. [paper] [code]
- [5] L. Pineda, T. Fan, M. Monge, S. Venkataraman, **P. Sodhi**, et al. *Theseus: A library for differentiable nonlinear optimization. Neural Information Processing Systems (NeurIPS) 2022*. [paper] [code]
- [6] **P. Sodhi**, E. Dexheimer, M. Mukadam, S. Anderson, M. Kaess. *LEO: Learning Energy-based Models in Factor Graph Optimization. Conference on Robot Learning (CoRL) 2021*. [paper] [code]

PATENTS

- [7] **P. Sodhi**, R. McDonald, S. Branavan, V. Cirik *Automation of Tasks using Language Model Prompts*
Patent No. 18/342,350, Filed: Jun 27, 2023, Pending.
- [8] S. Ray, K. Weinberger, S. Branavan, **P. Sodhi** *Automated Key-Value Extraction using Natural Language Intents*
Patent No. US3456789C1, Filed: Dec 9, 2022, Pending.

AWARDS

- **FRAIM Research Scholarship**, Selected as 1 in 7 students from CMU School of Computer Science.
By Facebook AI Research, USA · Mar 2022
- **University Presidential Fellowship** Towards excellence and impactful research by Ph.D. students.
By Carnegie Mellon University, USA · Oct 2018
- **Best Conference Paper Finalist** Top 0.25% of accepted papers.
By IEEE Conference on Intelligent Robots and Systems (IROS) · Sep 2021
- **Best Application Paper Finalist** Top 0.4% of accepted papers.
By IEEE Conference on Intelligent Robots and Systems (IROS) · Sep 2017